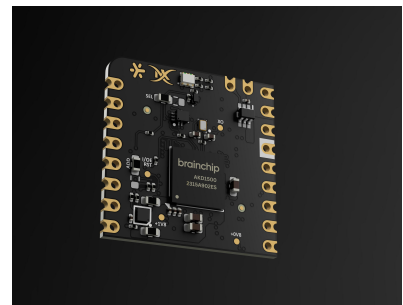


BrainBoard 15 Datasheet

User Manual

SKU: BB15-0001



Description

Nicla style embeded module integrating the Brainchip AKD1500 Neuromorphic AI accelerator

Target Areas:

Embedded AI, prototyping, edge vision

CONTENTS

1	Application Examples	3
2	Features	3
2.1	General Specifications Overview	3
2.2	Accessories	3
2.3	Related Products	3
3	Ratings	3
3.1	Recommended Operating Conditions	3
4	Functional Overview	4
4.1	Pinout	4
4.2	Block Diagram	4
4.3	Power Supply	4
5	Device Operation	4
5.1	Getting Started	4
5.2	Libraries and Firmware	4
5.3	Online Resources	4
6	Mechanical Information	4
6.1	Board Dimensions	4
6.2	Connectors	5
6.3	Peripherals and Actuators	5
7	Product Compliance	5
7.1	Compliance Summary	5
7.2	Regional Statements	5
8	Company Information	5
9	Reference Documentation	5
10	Document Revision History	5

1 Application Examples

Summarize the main use cases here. The reference document uses short introductory paragraphs followed by compact bullet lists.

- **Industrial automation:** Describe the primary industrial workflow or deployment.
- **Prototyping:** Explain how the board accelerates proof-of-concept work.
- **Edge AI:** Note the inference or sensing role in the wider system.

2 Features

2.1 General Specifications Overview

This section is usually a short paragraph plus a two-column summary table.

Feature	Description
Processor	Insert the MCU, accelerator, or SoC details here.
Memory	Flash, RAM, external storage, and relevant limits.
Connectivity	SPI, I2C, UART, USB, wireless, or other interfaces.
Power Supply	Supported voltage range and main power-input path.
Sensors	List onboard sensors and the purpose of each one.
Dimensions	Physical size in millimeters.
Weight	Board weight in grams.

2.2 Accessories

- Accessory or cable included with the product.
- Optional connector, header, or mechanical add-on.

2.3 Related Products

- Related base board or carrier.
- Related accessory or expansion module.
- Related development kit.

3 Ratings

3.1 Recommended Operating Conditions

Use a compact operating-range table for voltages, temperatures, and any required timing limits.

Parameter	Symbol	Min	Typ	Max	Unit
Input Voltage	V_{IN}	3.3	5.0	5.5	V
Operating Temperature	T_{OP}	-20	25	70	°C
Current Draw	I_{SYS}	–	120	250	mA

Power Tip

Use short note boxes for practical guidance. The reference PDF uses these to call out power and safety details without interrupting the flow of the section.

Safety Note

Document any voltage-level caveats, thermal constraints, or handling requirements here.

4 Functional Overview

4.1 Pinout

Place the pin summary here. If you have a diagram, insert it with `\includegraphics`.

4.2 Block Diagram

Add the architecture diagram or subsystem overview here.

4.3 Power Supply

Explain power domains, regulators, and any always-on versus switchable rails.

5 Device Operation

5.1 Getting Started

Link the quick-start path here: setup, flashing, examples, and first validation steps.

5.2 Libraries and Firmware

List the core software package, repository URL, and any version constraints.

5.3 Online Resources

- Documentation portal
- Library reference
- Example repository

6 Mechanical Information

6.1 Board Dimensions

Add the mechanical drawing or a dimensions table here.

6.2 Connectors

Describe connector types, mating parts, and mounting guidance.

6.3 Peripherals and Actuators

Summarize LEDs, buttons, microphones, cameras, or other user-facing hardware.

7 Product Compliance

7.1 Compliance Summary

CE	European Union
RoHS	Restriction of Hazardous Substances
REACH	Registration, Evaluation, Authorization and Restriction of Chemicals

7.2 Regional Statements

Reserve this space for declarations, classifications, and labelling notes.

8 Company Information

Company Information	Details
Company Name	BrainBoard Technologies
Company Address	Replace with your legal or support address.

9 Reference Documentation

No.	Reference	Link
1	Main documentation	https://example.com/docs
2	Library repository	https://example.com/library
3	Example projects	https://example.com/examples

10 Document Revision History

Date	Revision	Changes
17/06/2026	1	Initial template release.